TM Forum Component

Anomaly Management

TMFC041

Maturity Level: **Alpha**	Team Approved Date: 26-Nov-2024
Release Status: Pre-production	Approval Status: Member Evaluated
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Notice

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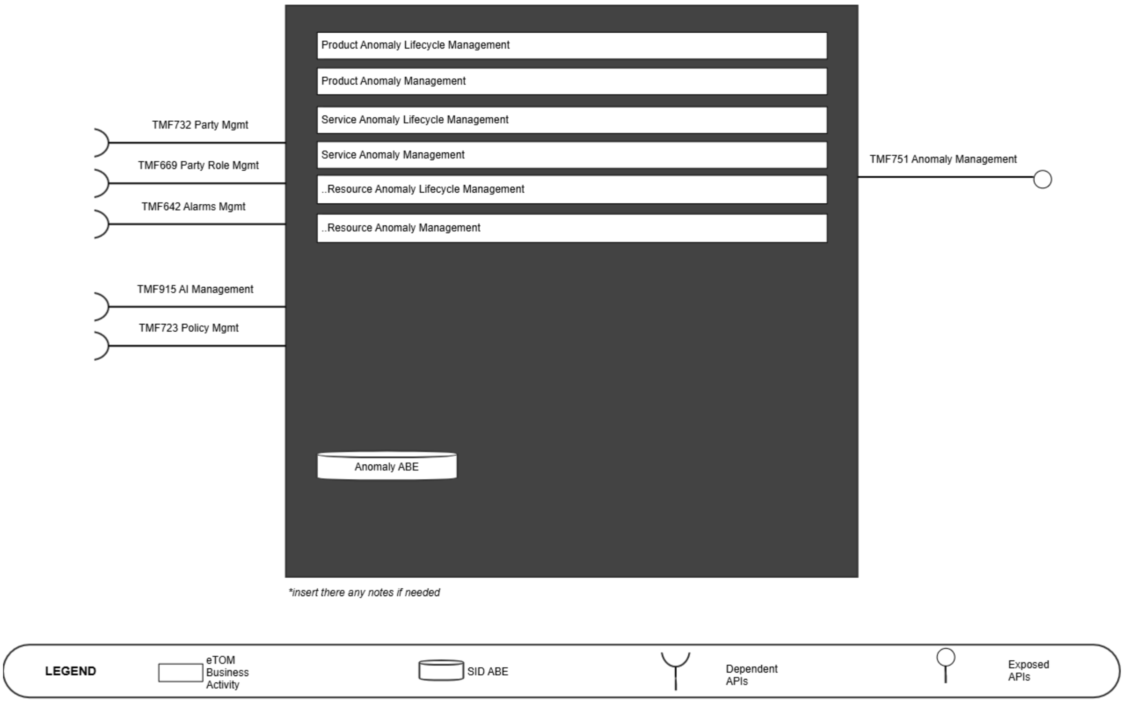
1. Overview

1. TAC-208 IG1171 (update) Component Definition to v4.0.0 and incorporate IG1245 Principles to Define ODA Components
2. [TAC-250] IG 1171 Improvements Some observations & recommendations. - TM Forum JIRA
3. [TAC-214] Interface Standardization needs all 3 stages of process to be developed - TM Forum JIRA
4. [TAC-226] Overview - TM Forum JIRA
5. ODA-846 Summary of ODA component Template enhancements for 14th Sep Review

Component Name	ID	Description	ODA Function Block
Anomaly Management (Anomaly Detection)	*TMFC041*	The Anomaly Management ODA Component is responsible for predicting, detecting and managing anomalous patterns in the usage and behavior of products, services and resources in order to avoid occurrence of problems. Anomaly Management ODA component has functionality to predict, detect and flag unusual patterns or outliers for products, services, and resource entities, and initiating mitigatory actions to prevent outlier patterns in usage or behavior to become problems. In the event of failure to predict or detect an early warning sign to a possible problem occurrence, the component hands of mitigatory actions to the ODA problem management component. Anomaly Management component enables define anomaly specifications based on given / known "normal" states or specifications of product/service and/or resource usage or behavior. *Note: In this release, this ODA Component is specified with functionality and APIs to detect anomalies based on anomaly specifications for product, service and / or resource. It includes the functionality to manage these specifications, administering the lifecycle of anomaly specification (for products/service and resource), and describing relationships between anomaly specifications. It also provides reporting on product/service/resource anomaly specifications and their changes, and facilitates easy and systematic indexing and*	*Intelligence Management*

Component Name	ID	Description	ODA Function Block
		access to specified and observed anomalous patterns - with manual or automatic anomaly management specification capture.	

TMFC041 : Anomaly Management (Detection)



2. eTOM Processes, SID Data Entities and Functional Framework Functions

2.1. eTOM business activities

<Note to not be inserted onto ODA Component specifications: If a new ABE is required, but it does not yet exist in SID. Then you should include a textual description of the new ABE, and it should be clearly noted that this ABE does not yet exist. In addition a Jira epic should be raised to request the new ABE is added to SID, and the SID team should be consulted. Finally, a decision is required on the feasibility of the component without this ABE. If the ABE is critical then the component specification should not be published until the ABE issue has been resolved. Alternatively if the ABE is not critical, then the specification could continue to publication. The result of this decision should be clearly recorded.>

eTOM business activities this ODA Component is responsible for.

Identifier	Level	Business Process Name	Description
1.2.24	*L2*	*Product Anomaly Lifecycle Management*	Product Anomaly Lifecycle Management business process establishes and controls all activities that are involved in overseeing, directing, administering, controlling and organizing the definition, detection/prediction, mitigation and learnings related to Product Anomaly Management.
1.2.24.6	*L3*	*Manage Product Anomaly Optimization*	Manage Product Anomaly Optimization business activity is in charge of actions that make the best and most effective use of Product Anomaly Management activities to improve Product Anomaly Management.
1.2.25	*L2*	*Product Anomaly Management*	Product Anomaly Management business processes establish actions that predict and detect aberrations or outlier events/activities, assess them for their impact, mitigate them, and record them before they ever become product problem management concerns.
1.2.25.2	*L3*	Detect Product Anomaly	Detect Product Anomaly business activity identify Product (buy/use/care) activities that are a deviation (aberrations or abnormal actions/events) from well-define norms or expectations.
1.4.17	*L2*	Service Anomaly Lifecycle Management	Service Anomaly Lifecycle Management business process establishes and controls all activities that are involved in overseeing, directing, administering, controlling and organizing the definition, detection/prediction,

Identifier	Level	Business Process Name	Description
			mitigation and learnings related to Service Anomaly Management.
1.4.17.1	*L3*	Manage Service Anomaly Definition	Manage Service Anomaly Definition business activity is in charge of defining and describing the information about data which is used in Service Anomaly Management.
1.4.17.6	*L3*	Manage Service Anomaly Optimization	Manage Service Anomaly Optimization business activity is in charge of actions that make the best and most effective use of Service Anomaly Management activities to improve Service Anomaly Management.
1.4.18	*L2*	Service Anomaly Management	Service Anomaly Management business processes establish actions that predict and detect aberrations or outlier events/activities, assess them for their impact, mitigate them, and record them before they ever become Service Problem Management concerns.
1.4.18.2	*L3*	Detect Service Anomaly	Detect Service Anomaly business activity identify Service delivery activities that are a deviation (aberrations or abnormal actions/events) from well-define norms or expectations.
1.5.20	*L2*	Resource Anomaly Lifecycle Management	Resource Anomaly Lifecycle Management business process establishes and controls all activities that are involved in overseeing, directing, administering, controlling and organizing the definition, detection/prediction, mitigation and learnings related to Resource Anomaly Management.
1.5.20.1	*L3*	Manage Resource Anomaly Definition	Manage Resource Anomaly Definition business activity is in charge of defining and describing the information about data which is used in Resource Anomaly Management.
1.5.21	*L2*	Resource Anomaly Management	Resource Anomaly Management business processes establish actions that predict and detect aberrations or outlier events/activities, assess them for their impact, mitigate them, and record them before they ever become Resource Problem Management concerns.
1.5.21.2	*L3*	Detect Resource Anomaly	Detect Resource Anomaly business activity identify Resource delivery activities that are a deviation (aberrations or abnormal actions/events) from well-define norms or expectations.

2.2. SID ABEs

<Note not to be inserted into ODA Component specifications: If a new ABE is required, but it does not yet exist in SID. Then you should include a textual description of the new ABE, and it should be clearly noted that this ABE does not yet exist. In addition a Jira epic should be raised to request the new ABE is added to SID, and the SID team should be consulted. Finally, a decision is required on the feasibility of the component without this ABE. If the ABE is critical then the component specification should not be published until the ABE issue has been resolved. Alternatively if the ABE is not critical, then the specification could continue to publication. The result of this decision should be clearly recorded.>

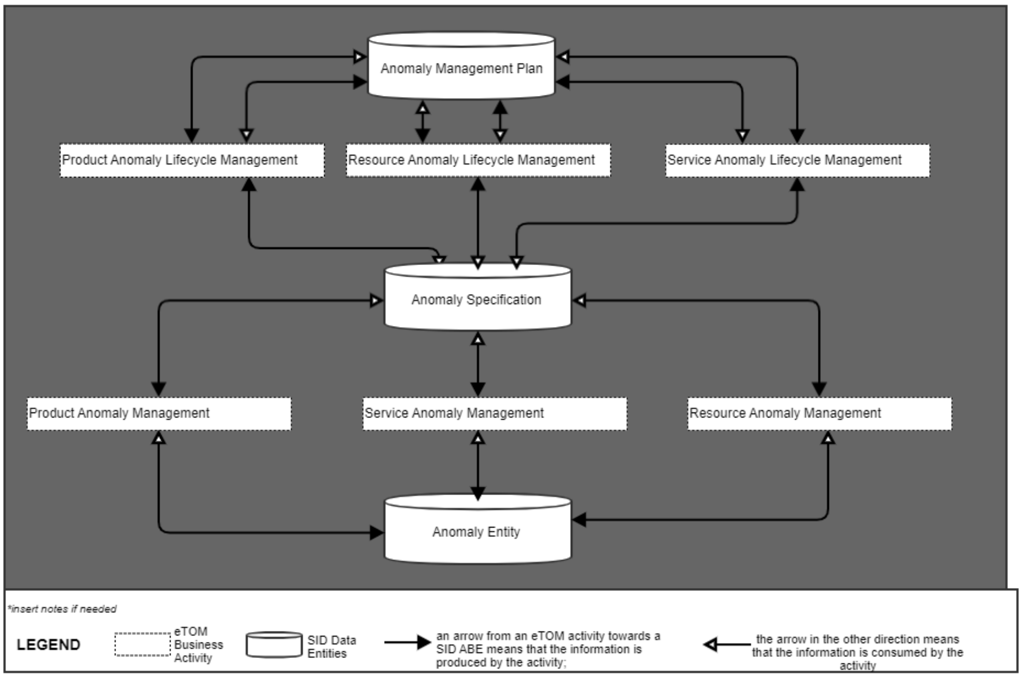
SID ABEs this ODA Component is responsible for:

SID ABE Level 1	**SID ABE Level 2 (or set of BEs)***
Anomaly ABE	Anomaly Entity ABE
Anomaly ABE	*Anomaly Management Plan ABE*
Anomaly ABE	*Anomaly Specification ABE*

*: if SID ABE Level 2 is not specified this means that all the L2 business entities must be implemented, else the L2 SID ABE Level is specified.

2.3. eTOM L2 - SID ABEs links

eTOM L2 vS SID ABEs links for this ODA Component.



2.4. Functional Framework Functions

Function ID	Function Name	Function Description	Aggregate Function Level 1	Aggregate Function Level 2
1352	Anomaly Pattern Collection	This function collects and stores "patterns" from various sources for use in anomaly detection. It takes input from a pattern discovery or pattern management source to be used internally for anomaly pattern processing	Anomaly Management	**Anomaly Detection**
1353	Anomaly Normal Description Capture	Anomaly Normal-Day Description Capture function is used to capture reference points to be used in the anomaly pattern processing. It takes as inputs of "normal-day" behavior, trend or pattern etc. and outputs a categorization of patterns	Anomaly Management	Anomaly Detection
1354	Anomaly Pattern Correlation	This function exposes analysis of the correlation between different anomaly patterns and their possible causes and impacts. It takes as input a pattern dataset and outputs the correlations between different anomaly patterns	Anomaly Management	Anomaly Detection
1355	Anomaly Learning	This function uses training methods to train and update the anomaly detection models based on data patterns and feedback (man/machine). It takes as input a dataset and outputs a model that can identify anomalies	Anomaly Management	Anomaly Detection
1356	Anomaly Alerting	This function sends alerts out (to users or other components) when an anomaly is detected. It takes as input the data and the model from the Anomaly Learning function, and outputs a signal to alert when the model identifies anomalies	Anomaly Management	Anomaly Detection
1357	Anomaly Reporting	This function generates and renders reports on the detected anomalies and their details. It takes as input the data and the anomalies identified, and outputs a report detailing these anomalies	Anomaly Management	Anomaly Detection
1358	Anomaly Pattern Analysis	This function performs statistical and graphical analysis on patterns received and inferred to identify trends and outliers. It takes as input patterns and outputs an analysis of the patterns found to be outliers	Anomaly Management	Anomaly Detection

Function ID	Function Name	Function Description	Aggregate Function Level 1	Aggregate Function Level 2
1359	Univariate Anomaly Detection	Detects anomalies in one variable, like revenue, cost, etc. The model is selected automatically based on pattern. It takes as input a univariate dataset and outputs the identified anomalies	Anomaly Management	Anomaly Detection
1360	Multivariate Anomaly Detection	Detects anomalies in multiple variables with correlations, which are usually gathered from products, services or resources (including devices, IoT and equipment etc.) or other complex systems. It takes as input a multivariate dataset and outputs the identified anomalies	Anomaly Management	Anomaly Detection
1361	Anomaly Model Selection	This function selects the best-fitting anomaly detection model from own or an external model repository for a given data set or scenario. It takes as input a dataset and potential models, and outputs the best model for a given anomaly detection requirement	Anomaly Management	Anomaly Detection
1362	Real-time Model Inferencing	This function (a.k.a Streaming Detection) performs anomaly detection on real-time data streams using a given or selected model. It takes as input new data and the selected model, and outputs identified anomalies in real-time	Anomaly Management	Anomaly Detection
1363	Anomaly Batch Inferencing	This function performs anomaly detection on batch data sets using given or selected model. It takes as input a batch of new data and the selected model, and outputs identified anomalies	Anomaly Management	Anomaly Detection
1364	Anomaly Pattern Matching	This function compares the detected anomalies with known anomaly patterns to find similarities and differences. It will compare observed patterns to known patterns (either anomalous or non-anomalous) to identify matches. It takes input from a dataset and a set of known patterns, and output matches between the observed and known patterns	*Anomaly Management*	*Anomaly Detection*

3. TM Forum Open APIs & Events

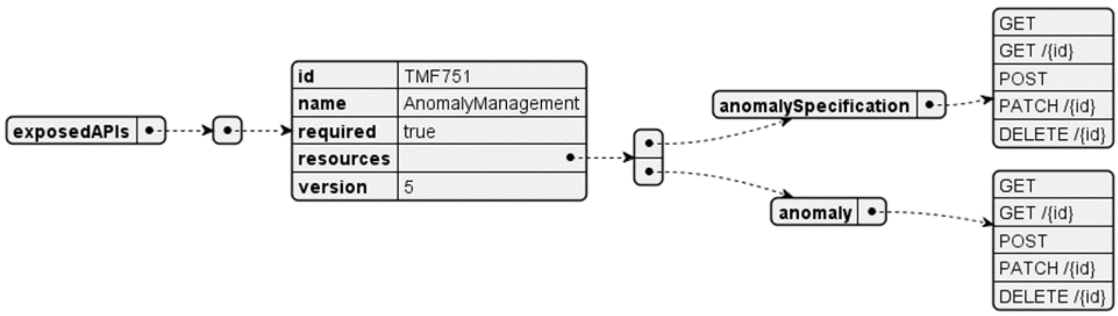
The following part covers the APIs and Events; This part is split in 3:

- List of **Exposed APIs** - This is the list of APIs available from this component.
- List of **Dependent APIs** - In order to satisfy the provided API, the component could require the usage of this set of required APIs.
- List of **Events** (**generated & consumed)** - The events which the component may generate is listed in this section along with a list of the events which it may consume. Since there is a possibility of multiple sources and receivers for each defined event.

<Note note to be inserted into ODA Component specifications: If a new Open API is required, but it does not yet exist. Then you should include a textual description of the new Open API, and it should be clearly noted that this Open API does not yet exist. In addition, a Jira epic should be raised to request the new Open API is added, and the Open API team should be consulted. Finally, a decision is required on the feasibility of the component without this Open API. If the Open API is critical then the component specification should not be published until the Open API issue has been resolved. Alternatively if the Open API is not critical, then the specification could continue to publication. The result of this decision should be clearly recorded.>

3.1. Exposed APIs

Following diagram illustrates API/Resource/Operation:

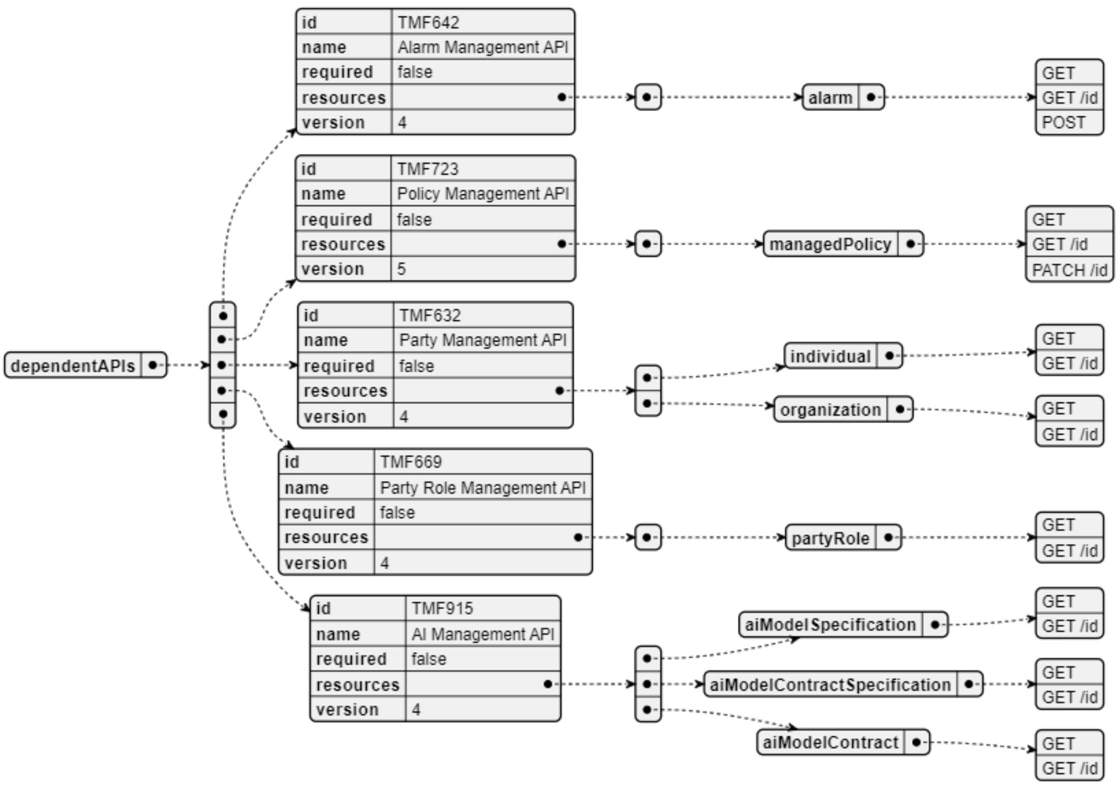


Note : TMF751 is in Pre-production state.

API ID	API Name	API Version	Mandatory / Optional
TMF751	*Anomaly Management*	*5*	*Mandatory*

3.2. Dependent APIs

Following diagram illustrates API/Resource/Operation potentially used by the Anomaly Management component:



API ID	API Name	API Version	Mandatory / Optional	Rationales	Resources
~~*TMF688*~~	~~*Event Management*~~	~~*4.0*~~	~~*Optional*~~	~~*For notification about detected Anomaly*~~	~~event:~~ ~~- GET~~ ~~- GET /{id}~~ ~~- POST~~
TMF642	*Alarm Management*	*4*	*Optional*	*Initiate Alarm as consequence of detected Anomaly*	alarm: - GET - GET /{id} - POST
TMF723	*Policy Management*	*5*	*Optional*	*Used for Rule based Anomaly Detection*	managedPolicy: - GET - GET /{id} - PATCH /{id}
TMF632	*Party Management*	*4*	*Optional*	*For managing the party associated with Anomaly*	individual: - GET - GET /{id} organization:

API ID	API Name	API Version	Mandatory / Optional	Rationales	Resources
				Management	- GET - GET /{id}
TMF669	*Party Role Management*	*4*	*Optional*	*For managing the Party Role Associated with Anomaly Management*	partyRole: - GET - GET /{id}
TMF915	*AI Management*	*4*	*Optional*	*For managing the AI Model interactions for Anomaly detection.*	aiModelSpecification: - GET - GET /{id} aiModelContractSpecification: - GET - GET /{id} aiModelContract: - GET - GET /{id} - POST

3.3. Events

The diagram illustrates the Events which the component may publish and the Events that the component may subscribe to and then may receive. Both lists are derived from the APIs listed in the preceding sections.


publishedEvent
subscribedEvent
AlarmManagement
AnomalyManagement
AIManagement
alarmCreateEvent
alarmStateChangeEvent
alarmDeleteEvent
anomalyCreateEvent
anomalyDeleteEvent
anomalyAttributeValueChangeEvent
anomalyStateChangeEvent
anomalySpecificationStatusChangeEvent
anomalySpecificationAttributeValueChangeEvent
anomalySpecificationCreateEvent
anomalySpecificationDeleteEvent
aiContractSpecificationCreateEvent
aiContractSpecificationDeleteEvent
aiContractSpecificationAttributeValueChangeEvent
aiModelSpecificationCreateEvent
aiModelSpecificationDeleteEvent
aiModelSpecificationAttributeValueChangeEvent
aiModelCreateEvent
aiModelStateChangeEvent
aiModelDeleteEvent
aiModelAttributeValueChangeEvent
aiContractCreateEvent
aiContractDeleteEvent
aiContractAttributeValueChangeEvent
aiContractStateChangeEvent

4. Machine Readable Component Specification

Refer to the ODA Component table for the machine-readable component specification file for this component. Also attached here

5. References

5.1. TMF Standards related versions

Standard	Version(s)
SID	*24.0*
eTOM	*24.0*
Functional Framework	*24.0*

5.2. Jira References

N/A

5.3. Further resources

1. *IG1219 AI Closed Loop Automation - Anomaly Detection and Resolution*
2. *TR308 Anomaly Detector ODA Component Requirements v1.0.0 – TM Forum*
3. *TR309 Anomaly Predictor ODA Component Requirements v1.0.0 (tmforum.org)*
4. *TR310 Anomaly Mitigator ODA Component Requirements v1.0.0 (tmforum.org)*
5. *TR308A Anomaly Detector Business Process Scenarios v1.0.0 – TM Forum*
6. *TR309A Anomaly Predictor Business Process Scenarios v1.0.0 – TM Forum*
7. *TR310A Anomaly Mitigator Business Process Scenarios v1.0.0 – TM Forum*
8. **TR308E: Anomaly Detector Use Cases (WIP)**
9. **TMFSNNN: Anomaly Management - Product Ordering Anomaly Detection Detailed Use Case (WIP)**
10. IG1228: please refer to IG1228 for defined use cases with ODA components interactions.

6. Administrative Appendix

6.1. Document History

6.1.1. Version History

Version Number	Date Modified	Modified by:	Description of changes
1.0.0 Alpha	12 Aug 2024	Emmanuel A. Otchere Manoj Nair	Initial Release
	21 Oct 2024	Manoj Nair	Removed TMF672 User & Roles Management from the dependent API list as it is expected to be supported by Canvas Added the relevant resources and operations against each API to align with the component spec template
1.1.0	19 Nov 2024	Gaetano Biancardi	• TMF688, removed from the core specification, moved to supporting functions • API version, only major version to be specified
1.1.0	26 Nov 2024	Amaia White	Final edits prior to publication

6.1.2. Release History

Release Status	Date Modified	Modified by:	Description of changes
Pre-production	12 Aug 2024	Emmanuel A. Otchere Manoj Nair	Component description, reference ISA objects, associated APIs and events
Pre-production	14-Oct-2024	Adrienne Walcott	Updated to Member Evaluated status
Pre-production	26-Nov-2024	Amaia White	Updated to version 1.1.0
Pre-production	17-Feb-2025	Adrienne Walcott	Updated to Member Evaluated status

6.2. Acknowledgements

This document was prepared by the members of the TM Forum Component and Canvas project team:

Member	Company	Role*
Emmanuel A. Otchere	Huawei	Project Chair (AI-CLA), Author
Manoj Nair	Netcracker	Author
Gaetano Biancardi	Accenture	Additional Input

**Select from: Project Chair, Project Co-Chair, Author, Editor, Key Contributor, Additional Input, Reviewer*